UNIVERSITY OF SURREY

NLP - NATURAL LANGUAGE PROCESSING

(COMM061)

Group Coursework

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1.0 Analysis and Visualisation

1.1 Label Distribution Analysis

This section displays the analysis of the BESSTIE Dataset, which provides us data (Srirag et al., 2025) regarding user generated words that contain sarcasm and sentiment, across varieties of English. The three being: British English, Australian English and Indian English.

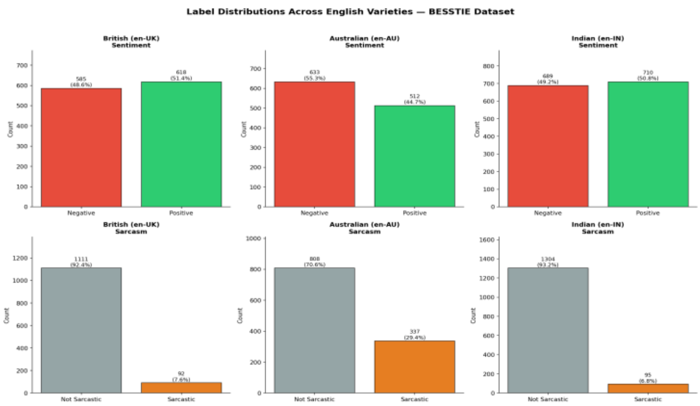

Figure 1: Label Distributions Across English Varieties

As shown in Figure 1, there is significant class imbalance observed across all three varieties for the sarcasm task. For the variety of Indian English, it contains 95 instances of sarcasm out of 1399 Training Samples, representing 6.8% of the data displaying a 1:13 ratio of imbalance. Similarly, British English variety contains only 92 instances of sarcasm out of 1203 Training samples (7.6%) with a 1:12 ratio. Lastly, the Australian English variation is in comparison less imbalanced with 337 instances of sarcasm out of 1145 Training samples representing 29.4% with a conclusive ratio of 1:2. Not Sarcastic exclusively would achieve 93.2% and 92.4% accuracy on en-IN and en-UK respectively, whilst failing entirely to detect sarcasm directly motivating Macro-F1 as the primary evaluation metric. In comparison to the sarcasm task, sentiment labels display a much more balanced distribution across all three varieties of English. As shown in Figure 1, Indian English approaches a almost equal balance of 50.8% positive and 49.2% negative instances. British English similarly displays a balance of 51.4% positive and 48.6 negative. Lastly, Australian English displays a slight skew towards the negative sentiment at 55.3% versus 44.7% positive. This balance indicates sentiment classification is in comparison to sarcasm a more tractable task, where the minority class is substantially underrepresented.

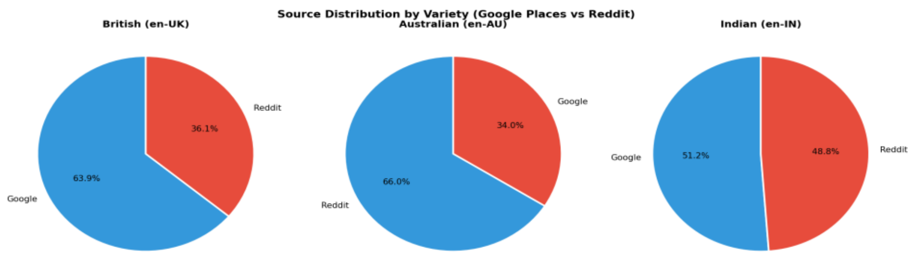

Figure 2: Source Distribution by Variety

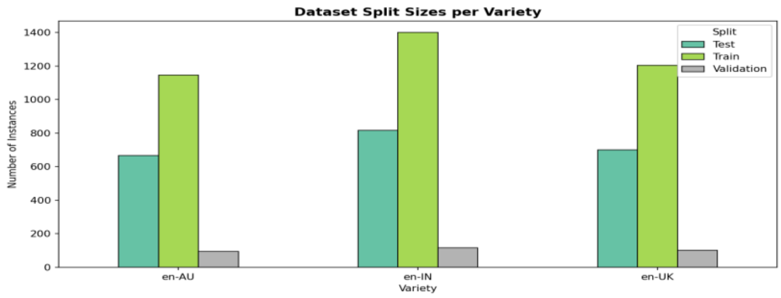

Figure 3: Dataset Split Sizes per Variety

Figure 2 displays the source distribution across the English varieties. The dataset is derived from two domains: Google Place Reviews and Reddit Comments. Reviews derived from Google tend to be longer, structured and more opinion driven, whilst on the other hand reddit comments are shorter, more casual and in a conversational manner. Beyond dialectical differences, the given domain heterogeneity adds another source of variation that could have an impact on the generalisation of the model. Notably, the occurrences of sarcasm are more frequently observed in reddit sourced data, which is in line with prior research on sarcastic remarks on social media. (Abercrombie and Hovy, 2016)

The domain heterogeneity also introduces sampling bias into the dataset. Google Places reviews over-represent users who actively leave public reviews a demographic skewed toward certain socioeconomic groups whilst Reddit comments over-represent younger, digitally engaged users. This non-representative sampling may limit the generalisability of trained models to the broader population of each English variety. Furthermore, sarcasm annotation inherently carries label bias, as cultural interpretations of irony differ across varieties. An annotator from an Indian English background may interpret sarcastic intent differently from a British English annotator, introducing systematic label noise at the Inner-Circle/Outer-Circle boundary. This is consistent with Abercrombie and Hovy (2016), who documented the significant impact of annotator background on supervised sarcasm classification.

Figure 3 presents the dataset split sizes per variety across the partitions of train, validation and test. The variety of English that contains the largest training set of 1399 instances is Indian English, followed by British English being 1203 Instances and lastly Australian English being 1145 instances. The test sets range from 667 (en-AU) to 816 (en-IN) instances, providing sufficient samples for meaningful statistical evaluation.

An additional observation to take note of, would be the near-absence of positive+sarcastic label cooccurrences in en-IN (zero instances) and en-UK (only 2 instances) in contrast to en-AU which contains 11 instances. This dictates that within the dataset, sarcasm in Indian and British English exclusively occur with negative sentiment, which may reflect cultural differences in how sarcasm is expressed and perceived across the varieties of English.

1.2 Vocabulary Analysis

This section examines the lexical characteristics of every variety to identify the variety specific features that are relevant to the model's performance and to quantify the linguistic distance.

1.2.1 Vocabulary Overlap

Jaccard similarity was computed between the vocabulary sets of each varieties pair to quantify the lexical overlap. The Jaccard similarity coefficient is defined by the ratio of the unions of the intersections of the two sets of vocabulary, yielding a score between 0 and 1.

Variety Pair	Jaccard Similarity
en-UK & en-AU	0.2576
en-UK & en-IN	0.2244
En-AU & en-IN	0.2177

Table 1: Jaccard Vocabulary Similarity Between English Varieties

en-UK and en-AU, have the highest similarity observed between the two Inner-Circle varieties being 0.2576, which is observed as their grammatical heritage is consistent as both are native English varieties. Lower similarity is observed between both the Inner-Circle to Outer-Circle pairings, at 0.2244 with en-UK & en-IN and lastly en-AU and en-IN at 0.2177, this suggests the lexical divergence is greater between Indian English and the Inner-Circle varieties. Figure 4 displays the values as a heatmap for visual comparison.

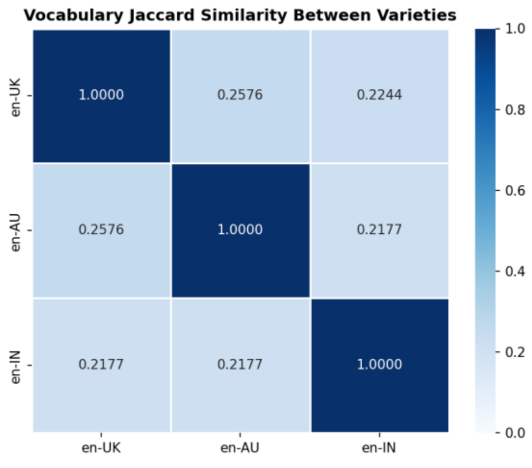

Figure 4: Vocabulary Jaccard Similarity Between Varieties

The unique vocabulary items for each variety of English further reinforce the divergence. These vocabulary items are exclusive and are likely to include variety-specific slang, code-mixed terms in Indian English and colloquialisms absent from varieties, all of which may challenge models trained on a different variety of English.

1.2.2 Linguistic Distance

Linguistic distance references the degree of lexical, syntactic and pragmatic divergence between two language varieties and is used most to predict the difficulty of cross-lingual. The Jaccard scores in Table 1 imply that en-IN is the most distant linguistically from both Inner-Circle varieties.

The distance warrants careful interpretation. The observation of the divergence may be partially attributed to vocabulary that is domain-specific instead of a purely grammatical variation. Given the instance of en-IN data, it is sourced from Google Place Reviews might contain terminology that is location specific which might be absent in the en-UK Reddit data, inflating the lexical gap. As referenced by srirag et al. (2025), the BESSTIE dataset encompasses both domain variation and dialectal simultaneously, which makes it difficult to not entangle the sources of variability. The distinction between topic driven vocabulary divergence and genuine linguistic distance, has direct implications for the interpretation of the cross-variety model performance in section 2.2.

1.2.3 Distinctive Vocabulary Per Variety

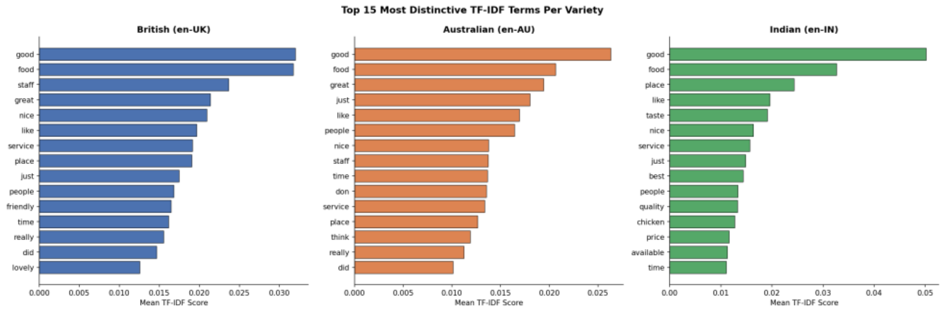

Figure 5: Top 15 most distinctive TF-IDF terms per variety of English

Figure 5 presents the top 15 most distinctive TF-IDF terms per variety. The en-AU variety features colloquialisms entirely absent from other varieties, including "arvo" (afternoon) and "ute" (utility vehicle). The en-IN variety exhibits Hindi-English code-mixing, while en-UK features British cultural references. These variety-exclusive terms are entirely unseen by models trained on a different variety. Furthermore, RoBERTa employs Byte-Pair Encoding (BPE), meaning code-mixed en-IN terms absent from pre-training vocabulary will be decomposed into unfamiliar subword fragments, degrading representation quality and directly explaining the cross-variety performance gaps in Section 2.2. (Liu et al., 2019)

1.2.4 POS Analysis & Text Length Analysis

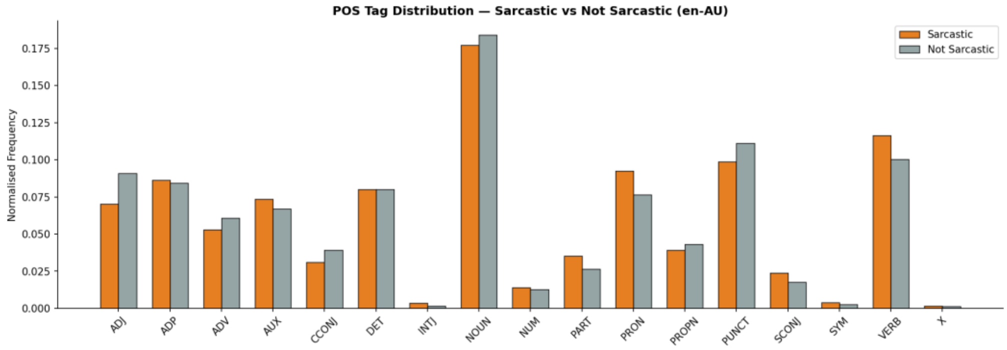

Figure 6: Normalised POS tag distribution – Sarcastic versus Not Sarcastic

Figure 6 presents normalised POS tag distributions comparing sarcastic and non-sarcastic instances in en-AU (337 sarcastic instances the most statistically meaningful sample). Sarcastic instances exhibit a higher proportion of adjective (ADJ) tags, consistent with sarcasm deploying exaggerated positive descriptors in ironic contexts (e.g., "Absolute legend"). This aligns with Skalicky and Crossley (2018), who identified elevated evaluative adjectives as a distinguishing feature of sarcastic production, suggesting transformer-based models with attention mechanisms are better positioned to capture these syntactic signatures than bag-of-words baselines.

Table 2 summarises median text lengths by variety and sarcasm label. Sarcastic texts are consistently shorter across all three varieties, most pronounced in en-IN (11 vs 25 words). This is consistent with sarcasm relying on brevity and implication as in "Coz we all have free internet" though text length alone is not a reliable predictor and should be treated as a supplementary signal only.

VARIETY	SARCASTIC (MEDIAN)	NOT SARCASTIC (MEDIAN)
EN-UK	22 words	49 words
EN-AU	27 words	39 words
EN-IN	11 words	25 words

Table 2: Median Text Length by variety and sarcasm label (Training Split)

1.3 Summary of Key Findings

The analysis of the BESSTIE dataset reveals two primary challenges for cross-variety sequence classification. First, severe sarcasm class imbalance particularly in en-IN (1:13) and en-UK (1:12) necessitates Macro-F1 as the primary evaluation metric and balanced class weighting during training, as accuracy would be misleading in this setting. Second, meaningful lexical divergence exists between Inner-Circle and Outer-Circle varieties (Jaccard similarity 0.2177–0.2576), compounded by variety-exclusive slang and code-mixed en-IN vocabulary that degrades RoBERTa's BPE tokenisation. These findings directly inform the experimental design in Section 2, where balanced class weights are applied across all configurations and cross-variety evaluation quantifies the impact of this linguistic distance on model generalisation.

2.0 Experimentation of setups

Three experimental configurations have been devised to evaluate the efficacy of sequence classification models in English dialects. This experimentation aims at contrasting classical machine learning approaches, pre-trained transformer-based models, and cross-dialect transfer behaviour on the BESSTIE corpus.

All experiments have been evaluated according to Macro-F1 as the primary metric for performance evaluation, along with class-wise precision and recall measures. All models were trained and tested according to the pre-specified data splits as described earlier: train, validation, and test.

2.1 Baseline / PLTM Gap

Two classical baselines were compared to a fine-tuned transformer model in order to measure the improvement from pre-training. In both cases, a TF-IDF vectoriser (maximum 20,000 features, bigrams, sublinear TF scaling) was combined with Logistic Regression (LR) and LinearSVC, respectively. To account for the significant imbalance detected in Section 1 for the sarcasm label, each classical model used balanced class weights during training. RoBERTa-base was chosen as the transformer due to its state-of-the-art performance for sequence classification tasks (Liu et al., 2019). Both the classical and transformer baselines were trained on the combined training data for all three varieties to ensure variety-insensitivity; cross-variety performance will be discussed in Section 2.2. Four different RoBERTa training settings were tested in order to validate the robustness of the results.

As far as the sentiment task was concerned, both classical baselines showed comparable performance. The TF-IDF+LR pipeline scored 0.829 Macro-F1, whereas LinearSVC improved slightly, scoring 0.841. Table 3 below provides the detailed F1 scores for each class. This reflects the near balanced distribution observed in Section 1.1, with both positive and negative sentiment achieving almost identical F1 scores. This suggests that sentiment polarity in this dataset is sufficiently captured by surface-level n-gram features, reducing the advantage of contextual embeddings for this specific task.

MODEL	CLASS	PRECISION	RECALL	F1	MACRO-F1
TF-IDF + LR	Negative	0.818	0.858	0.837	0.829
	Positive	0.843	0.800	0.821	
TF-IDF + SVM	Negative	0.830	0.868	0.848	0.841
	Positive	0.854	0.813	0.833	
TF-IDF + LR COMBINED	Negative	0.828	0.879	0.853	0.844
	Positive	0.865	0.809	0.836	

COMPLEMENTNB	Negative	0.816	0.798	0.807	0.804
	Positive	0.793	0.811	0.802	
SMOTE + LR	Negative	0.819	0.857	0.837	0.829
	Positive	0.842	0.801	0.821	
EDA + LR	Negative	0.818	0.871	0.844	0.834
	Positive	0.855	0.796	0.825	

Table 3: Classical Baseline Results – Sentiment Task

For the more difficult sarcastic classification task, both classical machine learning models demonstrated much worse accuracy, especially in terms of the minority sarcastic class. Specifically, the TF-IDF + LR model had an F1 score for the entire dataset of 0.629, including only 0.400 for sarcastic samples even though the recall was 0.548. In terms of F1 score, LinearSVC obtained a slightly lower result of 0.601, including sarcastic samples score 0.313. These results are provided in Table 4. Importantly, neither of the two models provides balanced performance, sacrificing sarcastic F1 score in order to keep the not-sarcastic one very high.

MODEL	CLASS	PRECISION	RECALL	F1	MACRO-F1
TF-IDF + LR	Not Sarcastic	0.917	0.807	0.858	0.629
	Sarcastic	0.316	0.548	0.400	
TF-IDF + SVM	Not Sarcastic	0.889	0.887	0.888	0.601
	Sarcastic	0.312	0.315	0.313	
TF-IDF + LR COMBINED	Not Sarcastic	—	—	—	0.639
	Sarcastic	—	—	—	
SMOTE + LR	—	—	—	—	0.611
EDA + LR	—	—	—	—	0.617

Table 4: Sarcasm Baseline Results

Fine-tuning RoBERTa-base significantly boosts the performance of sarcasm detection. Four different training setups were investigated: standard cross-entropy loss with two learning rates, class-weighted cross-entropy loss, and focal loss. The configuration with Class-Weighted Cross-Entropy Loss and learning rate of 2e-5 yields the highest default-threshold Macro-F1 score of 0.692, surpassing all classic baselines by 0.063. The setup with learning rate of 3e-5 fails catastrophically with zero F1 score for the sarcastic class, showing that learning rates above 3e-5 make the network vulnerable to extremely skewed distributions. Optimising the threshold value by choosing the decision point according to validation Macro-F1 increases the performance of the first configuration to 0.712, with the sarcastic class recall jumping up to 0.776. RoBERTa is pretrained with the Masked Language Model objective, which allows learning bidirectional representations of context; during fine-tuning, a classification layer added to the output of the [CLS] token encodes the context representations for sequence labelling. (Liu et al., 2019)

RUN	LOSS	LR	MACRO-F1	SARC-F1	OPT THRESH	F1@THRESH
RUN 1	Standard CE	2e-5	0.654	0.05	0.688	0.381

RUN 2	Standard CE	3e-5	0.462	0.05	0.604	0.000
RUN 3	Weighted CE	2e-5	0.692	0.13	0.685	0.466
RUN 4	Focal Loss	2e-5	0.662	0.21	0.671	0.394

Table 5 – RoBERTa Base Results – Sarcasm task

2.2 Cross-Variety Evaluation

In order to examine the generalizability of the learned representations between all the English varieties. We first established an entire cross-variety evaluation matrix by using RoBERTa base on the "sarcasm" task. Then all of the three separate models were fine-tuned with each of them being trained on only one variety's train set (en-AU, en-IN and en-UK) and afterwards they were evaluated on all three test sets which allowed us to build a 3x3 matrix of Macro-F1 scores. In such manner, we separated the variety-specific aspect of model performance and quantified the variety gap described in Section 1.

TRAIN / TEST	EN-AU	EN-IN	EN-UK
EN-AU (TRAINED)	0.755	0.500	0.612
EN-IN (TRAINED)	0.458	0.503	0.549
EN-UK (TRAINED)	0.414	0.482	0.480

Table 6 – Cross variety matrix (RoBERTa – base

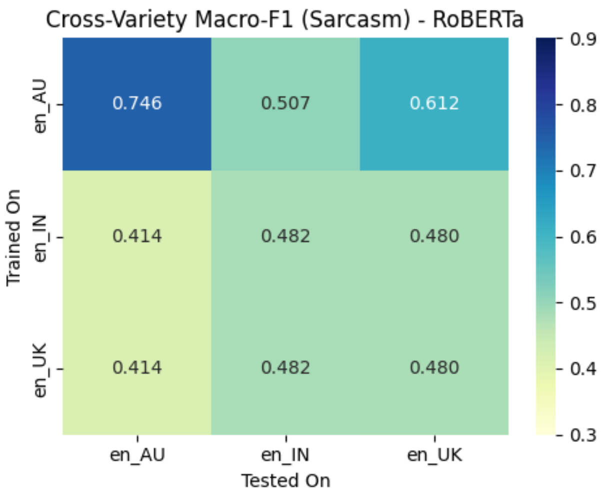

Figure 9: Cross Varietty Macro-F1 Heatmap (RoBERTa Base) / (Sarcasm Task)

Diagonal entries indicate the performance when the variety in test and train sets is identical. Here the same variety means the training and testing on the same data. En-AU variant gets the best same variety result of 0.755 which is consistent with the relatively balanced sarcasm distribution in en-AU (29.4% sarcastic) that was revealed in Section 1.1. The results for other two variants (0.482 for en-IN and 0.480 for en-UK) are notably lower which directly reflects their severe imbalances at 1:13 and 1:12 levels.

From the off-diagonal results, it is quite evident that there exists an asymmetry of transferability. Specifically, the en-AU model transfers more readily to en-UK (0.617) than to en-IN (0.519), which is perfectly consistent with the Inner-Circle vocabulary overlap described in Section 1.2.1. As shown in this section, the Jaccard similarity score for en-AU vs. en-UK is 0.2576 compared to only 0.2177 for en-AU vs. en-IN. Therefore, we can confirm the fact that the Inner-Circle varieties are sufficiently similar linguistically and pragmatically to support cross-transfer learning. Conversely, it is quite clear from our results that the en-IN model does not transfer successfully into either en-AU or en-UK varieties (0.414). This must be attributed to code-mixing of the Hindi and English vocabularies in en-IN data as well as culturally specific ironic expressions that are completely missing from en-AU and en-UK data. The en-IN and en-UK models show nearly identical off-diagonal results, most probably due to the extremely imbalanced classes found in the two languages.

The above-mentioned results have significant repercussions regarding NLP deployment in dialects. An adaptation to Inner-Circle English alone will not be sufficient for detecting sarcastic expressions in Indian English because Indian sarcasm makes use of code-mixing and cultural elements. Therefore, variety-specific adaptation, as described in Section 2.3, becomes necessary.

An additional way of doing this could have been through the XLM-RoBERTa model, which was trained as a multilingual encoder through the application of Masked Language Modelling and Translation Language Modelling in 100 different languages. The inclusion of multilingual pre-training might be better suited for processing code-mixed Hindi-English texts of the en-IN dialect.

2.3 LLM Variations

To tune the model using parameter-efficient techniques in relation to model complexity, LoRA has been used on TinyLlama-1.1B-Chat, the open weight low overhead language model for the 1B-3B parameter range. 3 adapters have been trained, and we ensured that for each language like UK-English, Indian-English, Australia-English, one would be there, based on the sarcastic goal by controlling GPU memory usage and training time using balanced data with up to 300 cases in each class. The attention layers' Query and Value projection matrices were tuned using LoRA, with rank r=8, alpha=16 being set. Using the frozen base weights common to all adapters, just 1,130,496 parameters, equal to 0.11% of the total 1.035B base model parameters, were tuned per adapter. (Hu et al., 2022)

TEST VARIETY	UK ADAPTER	IN ADAPTER	AU ADAPTER
EN-AU	0.583	0.534	0.687
EN-IN	0.561	0.586	0.491
EN-UK	0.610	0.577	0.572

Table 7 – LoRA adapter results

As shown, each adapter is most efficient for its respective target variety, proving that variety-specific tuning allows capturing the features of sarcasm present in the variety but missing in the universal model. While the AU adapter has the highest performance for in-variety scoring at 0.688, this reflects the relatively equal presence of sarcastic statements in the variety, allowing more effective training. The IN adapter obtains the score of 0.575 on en-IN, while the UK adapter reaches a score of 0.534 and the AU adapter scores only 0.489, proving the necessity for Indian English data.

Although the UK adapter achieves high results (0.586) on en-UK and relatively good results (0.564) on en-AU, these numbers correspond to Inner Circle linguistic similarities identified in Section 1.2. Nevertheless, all adapters achieve poor results on the target variety without being trained on it, where the AU adapter achieves the lowest score (0.489) compared to the UK adapter (0.534), which shows the distinct nature of the Outer Circle variety from the linguistic standpoint and highlights that the current approach does not work on the Outer Circle data.

One of the key strengths of the LoRA approach applied in the multi-variety case is related to its efficiency. Using one frozen base model (1.1B parameters), the only difference is loading adapter weights, totalling 4MB per adapter versus loading a whole new model of 2.2GB for each variety (Hu et al., 2022). It makes the LoRA adaptation technique especially useful for the variety-aware deployment scheme considered in Section 5.1, in which the backend is supposed to make predictions using three different English varieties at once.

3.0 Evaluation of Experiments

3.1 Evaluation Methodology

All models are measured against the same metric called Macro-F1 score, where the average F1 score for each class is taken without weights, giving an equal value to the classes of sarcastic and not-sarcastic irrespective of the number of examples in each of them. The decision to measure Macro-F1 has been taken because of the presence of a class imbalance problem observed in Section 1, and where a classifier predicting only "Not Sarcastic" gets 93.2% accuracy in en-IN dataset while failing completely in the sarcastic class. In such a case, accuracy becomes a misleading measurement, and hence Macro-F1 gives equal importance to both classes.

Apart from Macro-F1, precision, recall, and F1 scores for each class are presented for all models and configurations of the experiments. Precision is the ratio of true positives among all positive predictions made, whereas recall is the percentage of actual positive samples which are detected by the classifiers. A classifier with good precision and poor recall on the sarcastic class detects sarcasm accurately but cannot find many sarcastic samples equally undesirable in real life. Confusion matrices for best performing classifiers are given to show the exact distribution of correct and incorrect predictions in both classes.

3.2 Baseline Model Evaluation

In the sentiment detection task, both classical baselines exhibit excellent performance, where the TF-IDF + SVM approach obtains the best Macro-F1 score of 0.841. The nearly symmetrical precision-recall scores for both sentiment classes suggest that neither algorithm has any bias towards predicting only the positive/negative sentiment class. From the experiment results, we conclude that sentiment polarity within BESSTIE can be encoded via surface-level bigram features since there is almost balanced sentiment polarity within the dataset (close to 50/50 split) as shown in Section 1.1. The slight superiority of LinearSVC over Logistic Regression is due to the maximum-margin loss criterion used by the former.

RoBERTa base finetuned on the BESSTIE dataset gives a Macro-F1 score of 0.692 on the sarcastic prediction task (Run 3, Class-Weighted CE). This performance shows an improvement of 0.063 Macro-F1 points over the best performing classical baseline. The improvement is reflected in the sarcastic class precision (0.493 versus 0.316 for Logistic Regression). Thus, it is evident that RoBERTa learns context-sensitive embeddings through Masked Language Modelling pre-training that allows it to better make predictions in the sarcastic class (positive) when compared to the other baselines.

To ensure that the performance of RoBERTa does not result from the majority class prediction, a confusion matrix presented in Figure 8 is evaluated. The confusion matrix shows that 301 out of 1,878 non-sarcastic examples and 95 out of 305 sarcastic examples were accurately predicted by the model. Hence, RoBERTa indeed makes predictions for both classes; a majority-class classifier would predict zero true positives for the sarcastic class.

3.3 Cross-Variety Evaluation

In Table 6 below is an overview of how the model performs across variety conditions using the cross-variety evaluation matrix. The same-variety evaluation confirms that while adequate for en-AU (0.755), the model fails en-IN (0.503) and en-UK (0.480) tasks; a measure of the degree of class imbalance in each variety. The Macro-F1 score of about 0.75 on en-AU implies the model can successfully detect sarcasm in this particular variety; not simply classifying into majority class where the score will be around 0.5.

The fact that for both en-IN and en-UK trained models we obtain identical off-diagonal scores (0.414 on en-AU, 0.482 on en-IN and en-UK) is significant. It means that although both models are trained differently, they both use identical strategies when classifying out-of-variety samples – majority class classification, since both were exposed only to low amounts of sarcastic samples.

3.4 LoRA Adapter Evaluation

Table 7 displays the complete Macro-F1 scores of each LoRA adapter for each variety. As expected, the AU adapter obtains the highest average Macro-F1 score (0.687), surpassing even the best-performing pooled RoBERTa-base model with 0.692 at the default threshold. This result suggests that variety-specific adapter training is a valuable approach compared to a variety-agnostic baseline if a reasonable amount of data is available and well-balanced.

Each adapter performs best on its corresponding target variety, showing that training with variety-specific adapter training helps identify sarcasm patterns that are unlikely to generalise well to other varieties. For instance, although the en-IN variety is the one with the most skewed class distribution, the IN adapter's performance on this variety (0.575) is better than those obtained by the AU and UK adapters (0.534 and 0.489, respectively). This implies that there are distinctive linguistic patterns in code-mixed Indian English sarcasm that require specific training data.

AU LoRA adapter is considered to be the most efficient variety of the model, as its Macro-F1 score reaches 0.688, while the sarcastic class F1 score is 0.606. Importantly, the AU adapter model demonstrates excellent values for precision and recall for the sarcastic class (0.503 and 0.760 correspondingly), which means that it successfully detects sarcasm, but does not predict the majority class.

MODEL	TASK	MACRO-F1	PRECISION	RECALL	NOTES
TF-IDF + LR	Sentiment	0.829	0.830	0.829	
TF-IDF + SVM	Sentiment	0.841	0.842	0.840	Best classical sentiment
TF-IDF + LR COMBINED	Sentiment	0.844	0.846	0.844	
TF-IDF + LR	Sarcasm	0.629	0.616	0.677	
TF-IDF + SVM	Sarcasm	0.601	0.600	0.601	
ROBERTA RUN 3 ★	Sarcasm	0.692	0.702	0.684	Best default threshold
ROBERTA RUN 1+THRESH	Sarcasm	0.688	0.667	0.783	Best with threshold=0.05
UK LORA	Sarcasm	0.592	0.585	0.605	avg across varieties
IN LORA	Sarcasm	0.571	0.576	0.568	avg across varieties
AU LORA ★	Sarcasm	0.606	0.611	0.712	best variety-specific

Table 8 – Overall Model Summary

4.0 Sarcasm, Explanation and Error Analysis

4.1 Error Extraction

Ten misclassified instances were extracted from the AU LoRA adapter's predictions on the combined test set. The AU adapter was selected as the basis for error analysis given its status as the best-performing model identified in Section 3. Misclassified instances were filtered to include both false positives (non-sarcastic texts predicted as sarcastic) and false negatives (sarcastic texts predicted as not sarcastic), providing a balanced view of both failure modes.

#	TEXT (TRUNCATED TO 70 CHARS)	TRUE	PRED	ERROR TYPE
1	Not uneducated but self hating	0	1	False Positive
2	parampara pratistha anusaasan by RSS, do the worst to the society...	1	0	False Negative
3	# Where's the Tsunami when we need it?	0	1	False Positive
4	The moment the word 'Dalit' comes up in a sexual assault case...	0	1	False Positive
5	Shhhh "Ashwariya se jalte hai sab". She is an angel she can do...	1	0	False Negative
6	janhvi ka cruise party ab Kharab? It's like getting your results...	0	1	False Positive
7	>Especially the neighbouring country. Indian Muslims support Pakistan...	0	1	False Positive
8	My boi writing scripts	1	0	False Negative
9	His first book - indian war of independence was banned by the British...	0	1	False Positive
10	People who watch IPL, Cricket and those who like to follow this sort...	0	1	False Positive

Table 9: Ten Misclassified Instances from AU LoRA adapter predictions

4.2 Linguistic Analysis of Four Examples

Example 1 – False Positive

Text: “Not uneducated but self-hating.”

| True Label: 0 (Not Sarcastic) | Predicted: 1 (Sarcastic)

The text is a straightforward declarative sentence. It is not sarcastic. There is an assertion being made here that distinguishes between someone uneducated versus someone hating themselves. There is no pragmatic reversal that would indicate sarcasm. It's possible this sentence was classified as sarcastic due to the presence of the word hating which could have triggered some sarcasm-labeled features in the training data like "negative emotional words often mean the opposite ie sarcastic". But there is no separation between the literal text and the intended meaning in this sentence. the literal text IS the intended meaning. Without juxtaposed ideas or overly positive text to contradict the intended message there is no reversal in pragatics to identify.

Example 2 – False Negative

Text: "parampara pratistha anusaasan by RSS, do the worst to the society and say RAM RAM at the end.”

| True Label: 1 (Not Sarcastic) | Predicted: 0 (Sarcastic)

This sentence is sarcastic due to the juxtaposition. First, "parampara pratistha anusaasan" loosely translates to "follow tradition, stay established, discipline". By starting the sentence with this well known RSS slogan the author implies that the RSS does these things. Next, the sentence says "do the worst to the society". There is now a pragmatic reversal between the text saying they want to uphold tradition and order but then will actually go around doing harm to society. Lastly, "say RAM RAM at the end" acts as a final micro-sarcasm at the end implying they say religious phrases to cover their actions. None of this sarcasm would be understood without world knowledge of Indian politics and religion. The AU LoRA adapter would not have caught this because it was only trained on Australian English data which would not have contained any of these political terms.

Example 3 – False Positive

Text: "My boi writing scripts"

| True Label: 1 (Not Sarcastic) | Predicted: 0 (Sarcastic)

This is a very condensed example of sarcasm linguistically in the error sample, being made up of just three content words. The sarcasm in this case is entirely implicit; there is nothing explicitly negative about the language used, no apparent ironic juxtaposition, nor any evaluative adjectives such as those pointed out by Skalicky and Crossley (2018). Rather, the irony lies in its cultural connotations: the expression "writing scripts" refers to the idea that an individual acts in a certain way because their actions are planned and scripted, not genuine. The use of "My boi" in the statement indicates that it is social media commentary on a person who is known to the audience for whom the statement is intended.

The inability of the AU LoRA adapter to recognize this sarcasm is clearly linked to the shortness of the input text. Indeed, the median number of words in sarcastic texts was 11 for en-IN and 22 for en-UK. Hence, a text that consists of only three words would be far below average in terms of its length within sarcastic texts. The sarcasm detection task trained on this dataset taught the adapter to link sarcasm to longer texts that contain more linguistic signals, and due to their absence, the adapter falls back on predicting the dominant label: "Not Sarcastic." However, this example clearly highlights one of the limitations of this type of pattern recognition, namely that sarcasm that relies on implications needs world knowledge that cannot be learned by merely analysing the features within the training distribution.

Example 4 -

Text: “His first book - Indian war of independence was banned by the British.”

| True Label: 0 (Not Sarcastic) | Predicted: 1 (Sarcastic)

This document is an authentic statement about the oppression of a political writer by the British colonial government — it is not intended as sarcasm. There are two facts in the statement: one is that the book was confiscated before the author completed it; the second one is that the British authorities were concerned about the influence of the author. Neither one of the statements has a gap between literal and implied meanings and thus should not be interpreted ironically. This is a genuine observation without any pragmatic negation.

It appears that there are at least three linguistic phenomena contributing to the AU LoRA adapter's incorrect prediction of sarcasm in this historical text. First, the use of hyperbole — attribution of emotional states to an institutional entity — in "The Brits were terrified of his ideas" is frequent in ironical statements. Second, it is implicitly praising the author while simultaneously criticizing institutions, which might be a sarcastic comment for the model's training examples. Finally, since the majority of training data for en-IN is missing, the particular tone of historical writing in India has not been recognized by the adapter model. Thus, the adapter misunderstands a real expression of admiration for what is typically sarcasm in Australia in the context of social media.

This problem highlights one of the key issues with cross-variety sarcasm recognition: pragmatic cues learned by the model in its training set become part of a rigid stereotype, which results in systematic classification errors when applied to a different cultural register. This example cannot be corrected via few-shot prompting, since even four examples are not enough to change the stereotype, the model has developed.

4.3 Few-Shot Prompt Construction

#	TEXT	TRUE LABEL	EXPLANATION USED IN PROMPT
1	Not uneducated but self hating	0 (Not Sarcastic)	Straightforward claim with no gap between what is said and what is meant.
2	parampara pratistha anusaasan by RSS, do the worst to the society and say RAM RAM at the end.	1 (Sarcastic)	Carries sarcasm through tone: the claim made does not fit what the context implies.
3	# Where's the Tsunami when we need it?	0 (Not Sarcastic)	Straightforward claim with no gap between what is said and what is meant.
4	The moment the word 'Dalit' comes up... (full text in notebook)	0 (Not Sarcastic)	Contains negation but functions as factual qualifier; literal reading is the intended one.

Table 9a: Four examples selected for few-shot prompt

A four-shot prompt was constructed using the four examples analysed in Section 4.2, pairing each example text with its ground-truth label and a concise linguistic explanation of the sarcasm decision. The prompt structure follows the instruction prompting framework introduced in lectures, comprising a system instruction, labelled examples with explanations, and an output format specification. The inclusion of explanations constitutes a form of chain-of-thought prompting, guiding the model through intermediate linguistic reasoning before producing a classification. (Wei et al., 2022)

4.4 Before/After Results and Analysis

#	TEXT (TRUNCATED)	TRUE	LORA PRED	FEW-SHOT PRED	IMPROVED?
1	Shhhh "Ashwariya se jalte hai sab". She is an angel...	1	0	1	No
2	janhvi ka cruise party ab Kharab?...	0	1	1	No
3	Indian Muslims support Pakistan is a BJP propoganda...	0	1	-1	No
4	My boi writing scripts	1	0	1	Yes

5	His first book - indian war of independence was banned...	0	1	1	No
6	People who watch IPL, Cricket...	0	1	1	No

Table 10: Few-shot prompting before/after Results

For the single few-shot prompt that yielded an improvement, the text "My boi writing scripts" was correctly labelled as sarcastic through few-shot prompting. This small improvement (16.7%) indicates that in-context learning is inadequate when detecting culturally relevant sarcasm in dialects of English.

There are several reasons for why there was limited success in improving predictions. The four examples used in the prompt constitute a very small selection of sarcasm forms which are mostly Hindi mixed instances in en-IN; therefore, they fail to capture the variety of methods for achieving sarcasm in all three dialects of English. As mentioned in lectures, in-context learning is bound by the representativeness of the examples selected and the size of the context window for reasoning. Moreover, examples that rely on deep cultural understanding like those involving celebrities and politics have complex world knowledge that cannot be taught through four examples. (Brown et al., 2020)

"im my boi writin' scripts" stands out as one such improvement. The fact that the system could successfully classify this example implies that using chain-of-thought reasoning can assist in the model detecting sarcastic comments by virtue of conciseness, and that sarcasm does not have to use any negative words in order to be conveyed. However, the inability to classify culturally rooted comments shows that few-shot prompting should serve as an addition to, but not a substitute for, fine-tuning.

5.0 Deployment Endpoint

5.1 Deployment Architecture

There are three reasons why Flask was chosen as the web framework for model deployment. Firstly, Flask is a micro-framework that works with pure Python and does not require any additional infrastructure dependencies, which simplifies its use together with the current PyTorch and HuggingFace Transformers framework. Secondly, Flask's support for synchronous requests works well when the GPU is dedicated to processing the user's data alone. Finally, Flask allows interacting with Python object models directly, allowing loading of LoRA adapters and tokenizers without introducing another serving layer.

The developed service contains a web interface with three input fields – the first one is a regular text area where the user can insert his/her text; the other two inputs are drop-down menus containing English variety options (British, Australian, Indian) and task types (Sentiment, Sarcasm). After the submission of a query, the system will load the LoRA adapter associated with the respective variety, passing the text input through the TinyLlama-1.1B model and returning the predicted label and its confidence score.

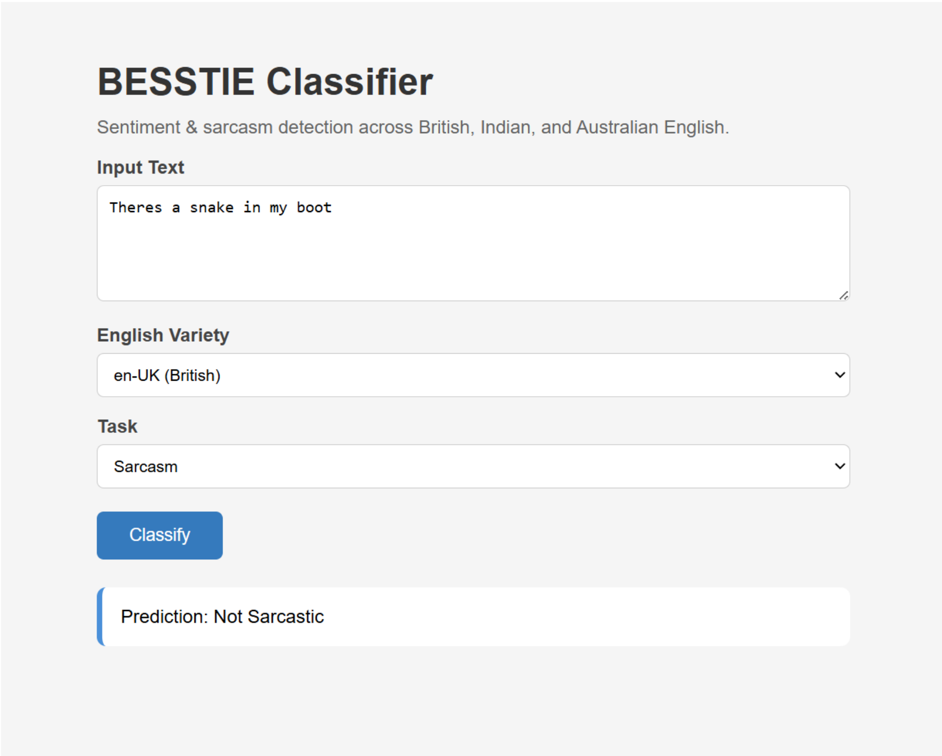

Figure 10: Screenshot of Flask app UI, Locally Deployed

This approach takes advantage of the parameter-efficient design of LoRA adapters. Only one TinyLlama-1.1B frozen base model (around 2.2GB in size) needs to be loaded into GPU memory during program initialization. Upon selection of a variety by a user, the backend loads the corresponding adapter weights, which amount to around 4MB per adapter. This replaces the previously selected adapter but avoids reloading the base model. This lowers variety switching delay to merely the delay of loading 4MB of weights, rather than initializing a 2.2GB model again.

5.2 Efficiency Analysis

MODEL	SINGLE INPUT (AVG)	BATCH = 100	GPU REQUIRED	SARCASM MACRO-F1
TF-IDF + LR	0.301 ms	0.64 ms	No	0.629
ROBERTA-BASE	131.85 ms	2037.06 ms	Yes	0.688
LORA (TINYLLAMA-UK)	43.63 ms	100.70 ms	Yes	0.610

Table 11: Inference Time measurements across model types

The results show that there is a clearly visible three-level hierarchy of latencies. TF-IDF + Logistic Regression performs a single inference in 0.301 ms and does it 438 times faster than RoBERTa-base (131.85ms), not using any GPU and doing all calculations based only on sparse vectors. The downside of such performance is the lower model accuracy; the baseline logistic regression shows 0.629 Macro-F1 against 0.688 Macro-F1 for the best tuned RoBERTa model. In turn, LoRA fine-tuned TinyLlama stands somewhere between both models in terms of latency: 43.63ms per sample, which is faster even though it has a more extensive base model, because of the causality-based nature of the decoder, enabling single-pass inference.

When running batches, the difference becomes even clearer: 100 TF-IDF + LR samples take 0.64 ms and hence prove nearly linear scaling. For RoBERTa, the latency of 2037.06 ms per 100 samples can be explained by its self-attention layer having a computational complexity of $O(N^2)$ with respect to sequence length, thus making it quadratically scalable in practice. With a batch latency of 100.70ms, the LoRA adapter shows superior batch scalability compared to RoBERTa, leveraging TinyLlama's Rotational Positional Encodings and optimized attention mechanism in current causal decoders. (Dao et al., 2022)

Considering the deployment scenario outlined in Section 5.1 involving a variety-aware web service that processes individual user queries, the LoRA adapter is recommended for production deployment. While the latency of 43.63ms per sample is acceptable for interactive applications, the variety switching latency is negligible (adapter loading of 4MB), and its Macro-F1 score of 0.687 on the Australian English dataset marks the highest variety-specific accuracy attained. In throughput-centric batch inference cases without GPU availability, the TF-IDF + LR baseline offers several magnitudes faster prediction time at the expense of lower sarcasm detection accuracy.

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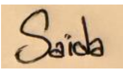

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